
BIOLOGY

9700/52

Paper 5 Planning, Analysis and Evaluation

May/June 2016

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Mark scheme abbreviations:

;	separates marking points
/	alternatives answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or extra guidance)
AW	alternative wording (where responses vary more than usual)
<u>underline</u>	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
ecf	error carried forward
I	ignore
mp	marking point (with relevant number)

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Question	Expected answer	Extra guidance	Mark
1 (a)	number of individuals or population of each type of/sort of/species present (in the sample) ; total number of individuals / all populations (of all species) ;	A count the number in different species A in context of any named organisms	[2]
(b)	<i>any 8 from:</i> 1 <i>ref. to</i> sampling in both areas/ grazed and ungrazed ; 2 any idea of marking out the area to be sampled ; 3 use a method of generating random numbers (to use co-ordinates) ; 4 use a (frame or point) <u>quadrat</u> (for individual samples) ; 5 place (quadrat AW) at coordinates ; 6 <i>ref. to</i> method of identifying or distinguishing different species/types/sorts of plant ; 7 <i>ref. to</i> counting/ recording of: number of individuals or the population of/ each type/sort/species present (in quadrat/plot) or the total number of all the plants present (in quadrat/plot) ; 8 same size quadrat/ same quadrat AW ; 9 same size plot in each area ; 10 same number of different quadrats/ samples per plot ; 11 replicate the procedure with a different plot in a given area ; 12 sample at different times of year/ seasons ;	I <i>any ref. to standardising environmental factors.</i> I if listed as the independent I <i>ref. to</i> transects e.g. tape measures/ use string and marker pole/ make a grid of plot e.g. random number generator/ app/ select number from a hat I throwing of quadrat <i>must be clear that the quadrat is the counting frame</i> <i>spelling of quadrat must be correct at least once</i> A descriptions, e.g. frame placed on the ground e.g. photographs/ key/ app/ expert/ nature guide/ AW A using letters or numbers for different species I percentage cover/ abundance scale e.g. 10 quadrats in each plot I repeat 3 times and find a mean A if only replicate with different plots in one area I repeat 3 times and take a mean I sampling on same day/ next week	

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Question	Expected answer	Extra guidance	Mark
	13 <i>safety</i> <i>any 1 from:</i> • <i>ref. to injury / getting lost and staying with group ;</i> • <i>allergy to plants and wearing gloves / protective clothing ;</i> • <i>allergy to pollen / hay fever and wearing mask or taking medication ;</i> • <i>ref. to uneven ground / hazardous plants or animals or environment and wearing suitable shoes / protective clothing ;</i>	I low risk A any suitable example – thorny / stinging plants, insect bites / stings, snakes, belligerent grazing animals and a suitable precaution	[max 8]
(c)	<i>independent:</i> grazed and / or ungrazed grassland and <i>dependent:</i> (mean) height (of plant) ;	A type of grass land I extent of grazing	[1]
(d) (i)	mode = 864 and median = 864 ;		[1]
(ii)	S_M grazed = 9.33 ; S_M ungrazed = 4.97 / 4.96 ;	<i>max 1 if answers are to 1 dp or 3 dp (9.3 / 9.329, 5.0 / 4.965)</i>	[2]
(iii)	860.1 ; to 879.9 ;	A ecf from 1(d)(ii) for correct calculation from incorrect S_M	[2]

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(iv)	any 2 from: 95% confident/sure/certain that the mean lies within these limits ; shows the reliability of the mean ; the ungrazed mean is more reliable (because it's smaller) ; the difference between means is significant because there is no overlap between CI for ungrazed and grazed ;	<i>must be a clear statement</i> R if ref. to accuracy or results AW ora the grazed is less reliable (because it is bigger)	[max 2]
(e)	any 2 from: sample from a large area ; <i>idea that</i> there is a long enough time interval, for marked individuals to mix into the population/between capture and recapture ; <i>idea that</i> the marking technique must not be toxic AW ; <i>idea that</i> the marking technique must not increase/decrease chances of survival ; marking technique must not fall off/be rubbed off/washed off animal ; <i>idea that</i> time is not so long that migration/life cycle changes (of the species) have occurred ;	I sample size I any specified times <i>need the idea of long enough for dispersal</i> e.g. increases or decreases chance of predation A in terms of inhibiting/changing movement or behaviour	[max 2]
(f)	ungrazed and because there are more seeds (to eat)/AW ;	A ungrazed as there will be larger plants and more places for inverts to hide from predators/protection from predators.	[1]
		Total:	[21]

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2 (a)	<i>any 3 from:</i> 1 body mass / weight ; 2 number of volunteers in each group ; 3 age of volunteers ; 4 no factor affecting air flow / lung capacity ; 5 (physical) fitness of volunteers ; 6 (type of) cigarette smoked ; 7 PEFR device / apparatus used ; 8 PEFR test done when volunteers are sitting down / standing up ; 9 time of day the PEFR test performed ; 10 ethnicity / race ;	I diet / sex / alcohol consumption / medication / drugs / range of number of packets of cigarettes ; A same number in each age group A asthma, CF, COPD, TB, lung cancer A disease affecting the lungs / breathing A living at altitude A minimum time since last cigarette I passive smoking A in terms of nicotine / tar / filter / brand A not after exercise / at rest	[max 3]

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Question	Expected answer	Extra guidance	Mark
(b)	any 3 from: support (max 2) conclusion 1 (an increase in the number of packets smoked decreases the PEFR measurement) 1 the <u>mean</u> PEFR decreases as the <u>mean</u> number of packets/cigarettes smoked increase ; 2 compare data from two PEFR and a trend on smoking or compare data from two number of packets smoked and a trend in PEFR ; 3 <u>highest</u> no. of packets/cigarettes smoked has the <u>lowest mean</u> PEFR ; conclusion 2 (the number of packets smoked increases with age) 4 as <u>mean</u> age increases the <u>mean</u> number of packets increases ; 5 compare data from two age groups and a trend on smoking or compare data from two mean number of packets smoked and a trend on age ; 6 <u>oldest</u> volunteers/group 5 smoked the <u>highest mean</u> number of packets ;	answers must either include both 'means' or link relevant data for any two groups (age or PEFR and number of packets smoked) from Table 2.1 I comparisons of age with PEFR must link PEFR values to the amount smoked/number of packets (not just quote from the table) e.g. (mean) PEFR decreases from 513.43 to 300.00 with increase in packets/cigarettes smoked e.g. (mean) PEFR decreases as the (mean) number packets increase from 0 to 189.22 A non-smokers/group 1 has the <u>highest mean</u> PEFR must link age values to the amount smoked/number of packets (not just quote from the table) must not use group 1 data here (26.42 and 0) e.g. (mean) number of packets increases from 30.61 to 189.22 with an increase in age e.g. (mean) age increases from 22.82 to 36.22 as the (mean) number of packets smoked increases A the <u>youngest</u> smokers/group 2 smoked the least <u>mean</u> number of packets A the <u>largest mean</u> number of packets was smoked by the <u>oldest</u> people	

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	does not support (max 2) conclusion 1 (an increase in the number of packets smoked decreases the PEFR measurement) 7 as the number packets increases and the values / range / standard deviation of PEFR of two of the groups overlap ; conclusion 2 (the number of packets smoked increases with age) 8 values / range / standard deviation of the ages (for each group) overlap or there are no distinct age groups / age groups overlap ; 9 group 2 smoke more packets than group 1 but (mean) age is lower ;	e.g. overlap between: group 1 / non-smokers and group 2 group 1 / non-smokers and group 3 group 2 and group 3 group 4 and group 5 A individuals in groups 1, 2, 3 and 4 all have a similar / same age	[max 3]
(c) (i)	there is no <u>significant</u> relationship / correlation between the decrease in the PEFR and the increase in the number of packets of cigarettes smoked or there is no <u>significant</u> decrease in the PEFR as the number of packets smoked increases or the increase in the number of packets smoked does not <u>significantly</u> decrease the PEFR ;	A there is no <u>significant</u> relationship / correlation between the increase in the number of packets of cigarettes smoked and the decrease in the PEFR	[max 1]
(ii)	any 2 from: number of volunteers small (est.) ; great(est) range in number of packets of cigarettes smoked (151–230) ; larg(est) standard deviation for number of packets of cigarettes ;	A has a range of 80 instead of 50	[max 2]
		Total:	[9]